G_! OF A QUASI-COHERENT MODULE ON THE LISSE-ETALE (FLAT-FPPF) SITE IS QUASI-COHERENT

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file stacks-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Functoriality of the lisse-etale morphism of topoi. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-lisse-etale-functorial. g_{i,!}(f'_i)^{-1} = f_i^{-1}g_!.
- Descent of quasi-coherence along coverings. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-check-qc-on-etale-covering. Reduces to the globally presented case.
- Structure sheaf of the lisse-etale site. Cohomology of Algebraic Stacks, Lemma stacks-cohomology-lemma-lisse-etale-structure-sheaf. g_!O = O_X.

Source reference. Original lemma 0788: sources/stacks-chow/tags.txt, source lines 9228–9228. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03DL: sources/stacks-chow/tags.txt, source lines 4424–4424. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03DE: sources/stacks-chow/tags.txt, source lines 4417–4417. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 06W1: sources/stacks-chow/tags.txt, source lines 8800–8800. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 07AT: sources/stacks-chow/tags.txt, source lines 9318–9318. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 07AZ: sources/stacks-chow/tags.txt, source lines 9324–9324. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0789: sources/stacks-chow/tags.txt, source lines 9229–9229. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0038: sources/stacks-chow/tags.txt, source lines 125–125. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:stacks-cohomology.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 078A: sources/stacks-chow/tags.txt, source lines 9230–9230. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0EM8: sources/stacks-chow/tags.txt, source lines 18298–18298. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $\mathcal{X}$ be an algebraic stack. Notation as in Lemma 0788.*

(1) *Let $\mathcal{H}$ be a quasi-coherent $\mathcal{O}_{\mathcal{X}_{lisse,étale}}$-module on the lisse-étale site of $\mathcal{X}$. Then $g_!\mathcal{H}$ is a quasi-coherent module on $\mathcal{X}$.*
(2) *Let $\mathcal{H}$ be a quasi-coherent $\mathcal{O}_{\mathcal{X}_{flat,fppf}}$-module on the flat-fppf site of $\mathcal{X}$. Then $g_!\mathcal{H}$ is a quasi-coherent module on $\mathcal{X}$.*

Proof. Pick a scheme U and a surjective smooth morphism $x : U \to \mathcal{X}$. By Modules on Sites, Definition 03DL there exists an étale (resp. fppf) covering $\{U_i \to U\}_{i\in I}$ such that each pullback $f_i^{-1}\mathcal{H}$ has a global presentation (see Modules on Sites, Definition 03DE). Here $f_i : U_i \to \mathcal{X}$ is the composition $U_i \to U \to \mathcal{X}$ which is a morphism of algebraic stacks. (Recall that the pullback "is" the restriction to $\mathcal{X}/f_i$, see Sheaves on Stacks, Definition 06W1 and the discussion following.) Since each f_i is smooth (resp. flat) by Lemma 07AT we see that $f_i^{-1}g_!\mathcal{H} = g_{i,!}(f_i')^{-1}\mathcal{H}$. Using Lemma 07AZ we reduce the statement of the lemma to the case where $\mathcal{H}$ has a global presentation. Say we have

$$\bigoplus\nolimits_{j\in J} \mathcal{O} \longrightarrow \bigoplus\nolimits_{i\in I} \mathcal{O} \longrightarrow \mathcal{H} \longrightarrow 0$$

of $\mathcal{O}$-modules where $\mathcal{O} = \mathcal{O}_{\mathcal{X}_{lisse,étale}}$ (resp. $\mathcal{O} = \mathcal{O}_{\mathcal{X}_{flat,fppf}}$). Since $g_!$ commutes with arbitrary colimits (as a left adjoint functor, see Lemma 0789 and Categories, Lemma 0038) we conclude that there exists an exact sequence

$$\bigoplus\nolimits_{j\in J} g_!\mathcal{O} \longrightarrow \bigoplus\nolimits_{i\in I} g_!\mathcal{O} \longrightarrow g_!\mathcal{H} \longrightarrow 0$$

Lemma 078A shows that $g_!\mathcal{O} = \mathcal{O}_{\mathcal{X}}$. In case (2) we are done. In case (1) we apply Sheaves on Stacks, Lemma 0EM8 to conclude. □